

Project Report

TriMart

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Table of Contents

1. Abstract
2. Introduction
3. Objectives of the Project
4. System Analysis
5. System Design
6. Implementation Details
7. Testing and Evaluation
8. Results and Discussion
9. Conclusion and Future Work
10. References

1. Abstract

The **TriMart** B2B Supplier Management System is an online system that we've created to ensure that business transactions between **TriMart** and its suppliers are sometimes seamless and transparent. The aim is to digitalise the entire life-cycle of supplied goods, from registration, product cataloguing, purchase order and payment, in one central place. By automating all that, the system eliminates the need for manual coordination, reduces paperwork and keeps everyone up to date and in real time. We used PHP for the backend, MySQL for database and (Tailwind) CSS/HTML for the frontend UI. Basically, **TriMart** can play the role of a retailer and directly chat with suppliers for the updating of the product, order placing, and monitoring the transaction. This work establishes a scalable digital ecosystem to increase supply chain efficiency and business collaboration.

2. Introduction

In today's competitive business environment retail organizations are heavily dependent on efficient and accurate coordination with suppliers. Traditional supplier management systems tend to be email, spreadsheet and telephone-based, resulting in delays, data replication and miscommunication. The **TriMart B2B Supplier Management System** is a contemporary and web-based solution that connects **TriMart** with its suppliers in a modernized digital position. The system permits the suppliers to register and maintain their company profiles, data upload and updating of products, and to receive purchase orders directly from **TriMart's** administration panel. By automating these processes, the system provides for improved transparency, speedier communication, and efficiency of operations. Both the **TriMart** administrators and the suppliers gain the advantages of less manpower and a better overview of business activities. This system represents the principles of a B2B digital marketplace, with a focus on partnership, collaboration and data-driven operations.

3. Objectives of the Project

1. The key goals of the **TriMart** B2B Supplier Management System are:
2. To overhaul and create a secure, web-based B2B platform that would link **TriMart** with multiple suppliers.
3. To automate the process of supplier registration, product management and order fulfilment.
4. In order to provide visibility of real-time inventory, orders and payment data to both parties.
5. To consolidate all supplier and product related data in a reliable and structured database.
6. To allow the administrators to manage the suppliers efficiently and keep track of the supplier's performance and transaction.
7. To minimise the potential for human error and ensure greater levels of accuracy in business-to-business activity.

8. To increase scalability and have a basis for further modules (logistics, invoicing, etc.).

4. System Analysis

4.1 Existing System

In many organizations, communication and management with suppliers is done in a manual way through offline methods such as phone, paper, and spreadsheet. This leads to great problems such as:

1. Slowness in communication and order processing.
2. Data loss and duplication are high possibilities.
3. Lack of centralized visibility among departments.
4. Difficulty in managing payments and deliveries.
5. Time consuming manual record maintenance.

Such issues cause problems in business productivity and bad supplier relationship.

4.2 Proposed System

The **TriMart** B2B Supplier Management System introduces a centralized, web-based solution of all activities related to the suppliers in a unified, web based platform.

Key Features:

1. Suppliers can create accounts, safely log in, and manage their profiles and the catalogs of their products.
2. **TriMart** administrators can view, approve or suspend supplier accounts and create purchase orders.
3. A centralized MySQL database is used to store all the transactions and supplier database.
4. The platform logs give real-time updates on stock, pending payments and order statuses.
5. The system provides data confidentiality using role based access control and session management.

This solution helps in improving the efficiency, transparency, and scalability in **TriMart's** supplier operations.

5. System Design

5.1 System Architecture

The entire system is designed and constructed as a three-tier B2B architecture:

1. **Presentation Layer (Frontend):**
We use HTML, (Tailwind) CSS to create an interactive and responsive web page that can be clicked by suppliers and admins.
2. **Application Layer (Backend):**
PHP is used to code server side. The engine is where all the work gets done - business logic, data crunching and such; communicating with the database in a secure manner.
3. **Database Layer:**
The structured data such as suppliers, products, orders and payments are all stored in a MySQL setup.

5.2 Database Design

The database of the **TriMart** B2B Supplier Management System supports all the roles (admins, distributors, suppliers, shippers). We opted to use relational with MySQL as it lends itself to keeping things tidy and future proof as well as making it easier to track and understand.

Primary entities and their function:

- **Companies** is where all the companies on the market, whether it are distributors, suppliers, or shippers, can be found. Each company record contains legal information: TIN, business licence, VAT etc. and the contact information.
- **Users** is where all the companies on the market, whether it are distributors, suppliers, or shippers, can be found. Each company record contains legal information: TIN, business licence, VAT etc. and the contact information.
- **Products:**
is a list of all products available. Global accessories are defined so that more than one supplier can attach their inventory to the same product record.
- **Supplier Inventory:**
links suppliers to their actual inventory of products, including price, stock level and current availability. Our search engine updates the table in real time and we never display out of stock articles.
- **Quotes and Quote Items:**
manage a back-and-forth between distributors and suppliers prior to an official order. Suppliers send out quotes to all distributors; distributors accept them and get offers, prices and validity dates.

- **Orders and Order Items:**
commands and order details between the distributor and the supplier. Each order will contain shipping & billing addresses, references to any quote information, and can be associated with a shipper once it has been assigned.
- **Transactions:**
records the financial aspect: type of payment (COD, bank or mobile banking), amounts and status
- **Delivery Tracking:**
the Delivery Tracking table is maintained by shippers and contains all the statuses of the shipment tracking entity (Assigned, InTransit, Delivered, Returned)
- **Connections:**
defines the direct relationships between distributors and suppliers; one distributor can make connection requests and create verified network.

Key Relationships

- One company can have several users.
- One supplier can be selling many products through the inventory table.
- Also, one distributor can send multiple quotes and make multiple orders to multiple suppliers.
- Each order can consist of many items and one or more transactions.
- One shipper is who makes several deliveries on orders.
- Connections are established one-to-one between the distributors and the suppliers they are working with.

The matrices maintain data integrity, have referential integrity and enable the B2B workflow with **TriMart's** administration responsibility.

5.3 User Interface Design

The **TriMart B2B System** provides a modern, responsive, and role-based user interface designed to simplify operations for all types of users. The frontend is built using **HTML, CSS (Tailwind CSS)**, ensuring a consistent look and smooth navigation across all devices.

Each user role has a dedicated portal that provides access to specific functions relevant to their responsibilities:

1. Admin Portal

The **Admin Portal** is the central control panel for **TriMart** management. It allows administrators to:

- Review and verify company registrations (Distributors, Suppliers, Shippers).
- Manage system-wide configurations and permissions.
- Monitor ongoing transactions, quotes, and orders.
- Approve or reject supplier connections and oversee payments.
- Generate analytical reports for system performance and trade statistics.

Admins have full visibility of marketplace activity and can intervene in case of disputes or irregularities.

2. Distributor Portal

Distributors act as **TriMart's** main clients who initiate business transactions. Their portal allows them to:

- Search and connect with verified suppliers.
- Request and review price quotes from multiple suppliers.
- Place purchase orders and monitor order fulfillment status.
- Track payments and manage order history.
- Communicate directly with suppliers and administrators through notifications.

The Distributor Portal is designed for clarity and speed, enabling efficient management of supplier relationships.

3. Supplier Portal

Suppliers use this interface to manage their products and handle requests from distributors. Core functions include:

- Managing the company profile and verification details.
- Adding, updating, and removing product listings.
- Responding to distributor quote requests and negotiating prices.
- Confirming or declining purchase orders.
- Monitoring payments and shipment progress.

The Supplier Portal emphasizes simplicity and control, allowing suppliers to maintain transparency and trust with their business partners.

4. Shipper Portal

Shippers handle the logistics and delivery of orders placed within the marketplace. Through their portal, they can:

- View assigned orders awaiting shipment.
- Update tracking numbers and delivery status in real-time.
- Confirm completed deliveries or report failed attempts.
- Communicate with distributors and suppliers about delivery schedules.

This module ensures end-to-end traceability of goods, improving the reliability of **TriMart's** supply chain.

User Experience and Accessibility

The interface for all roles follows a consistent design framework with:

- A **dashboard layout** for quick insights and key metrics.
- **Collapsible sidebars** for navigation.
- **Notification and alert systems** for updates.

6. Implementation Details

Programming Language: PHP

Frontend: HTML, CSS (Tailwind CSS),

Database: MySQL

Server Environment: XAMPP (Apache + MySQL)

Development Tools: Visual Studio Code, phpMyAdmin

Implementation Highlights:

- A single connection file `db_trimart.php` manages database access for all PHP modules.
- Session management ensures secure login for suppliers and administrators.
- Input validation and sanitization prevent SQL injection and cross-site scripting attacks.
- Modular coding structure allows independent development of login, registration, and dashboard components.
- The system architecture supports future scalability for more business features like automated invoicing or delivery tracking.

7. Testing and Evaluation

1. Unit Testing:

Each module (registration, login, order creation) was tested individually for functionality.

2. Integration Testing:

Verified data consistency between modules such as supplier management,

inventory, and payment systems.

3. **Functional Testing:**

Confirmed that all buttons, forms, and user interactions perform the correct actions.

4. **Database Testing:**

Ensured data is stored, retrieved, and updated correctly with no redundancy.

5. **Security Testing:**

Checked authentication, password protection, and access control between supplier and admin users.

Performance Testing:

Tested the system under simulated load with multiple concurrent supplier sessions.

8. Results and Discussion

The **Trimart B2B Supplier Management System** achieved its core goal of providing a digital link between **TriMart** and its suppliers.

Key Achievements:

- Streamlined communication and transactions between **TriMart** and suppliers.
- Automated supplier account creation and management.
- Centralized and secure storage of all business data.
- Faster processing of purchase orders and payments.

Discussion:

The system demonstrates the practical benefits of B2B digital transformation. It improves supply chain visibility, enhances data accuracy, and builds long-term business relationships. Although some backend modules (such as advanced analytics and payment APIs) are still in development, the existing implementation provides a solid foundation for expansion.

9. Conclusion and Future Work

The **TriMart B2B Supplier Management System** provides a scalable and efficient platform for managing supplier relationships in a business-to-business setting. It bridges the gap between **TriMart** and its suppliers by automating routine operations, improving accuracy, and enhancing transparency.

Future Enhancements:

1. Integration of payment gateways for automated billing and invoice management.
2. Addition of a reporting dashboard for sales analytics and supplier performance tracking.
3. Implementation of a notification system (email/SMS) for order updates and payment alerts.
4. Development of a mobile-friendly or native mobile app for suppliers.
5. Integration of AI-based analytics for forecasting and supplier behavior analysis.

Once fully completed, the system will serve as a complete digital B2B ecosystem for **TriMart's** supplier operations.

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